

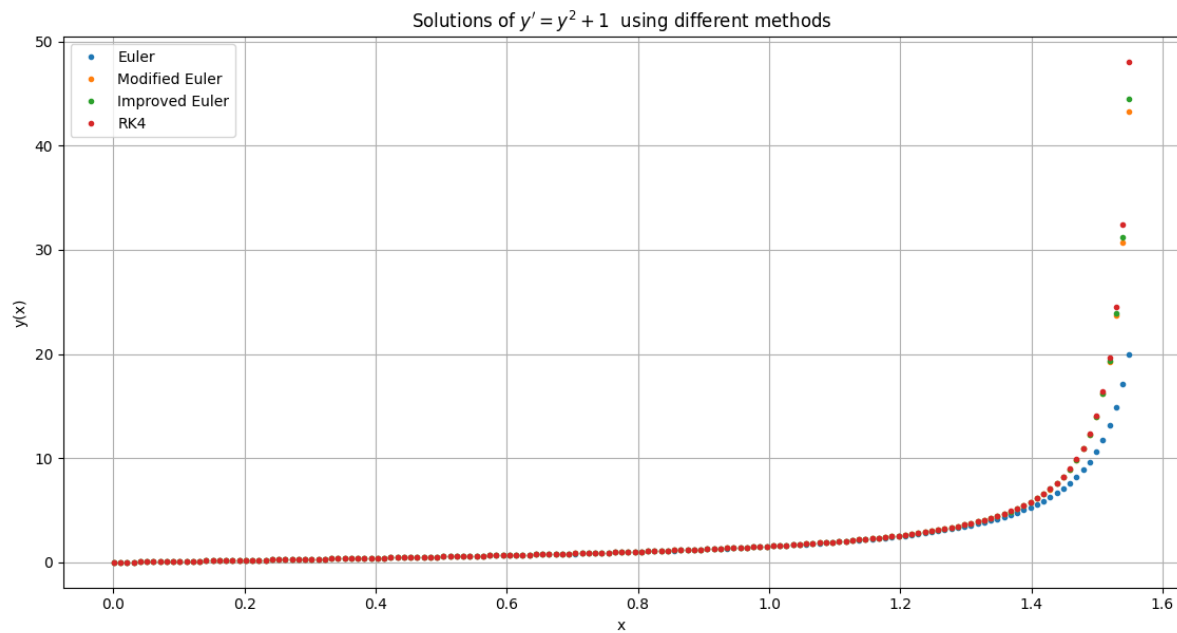
Q1:

Error in Euler's method: 28.067584

Error in Modified Euler's method: 4.768516

Error in Improved Euler's method: 3.597160

Error in RK4: 0.028337

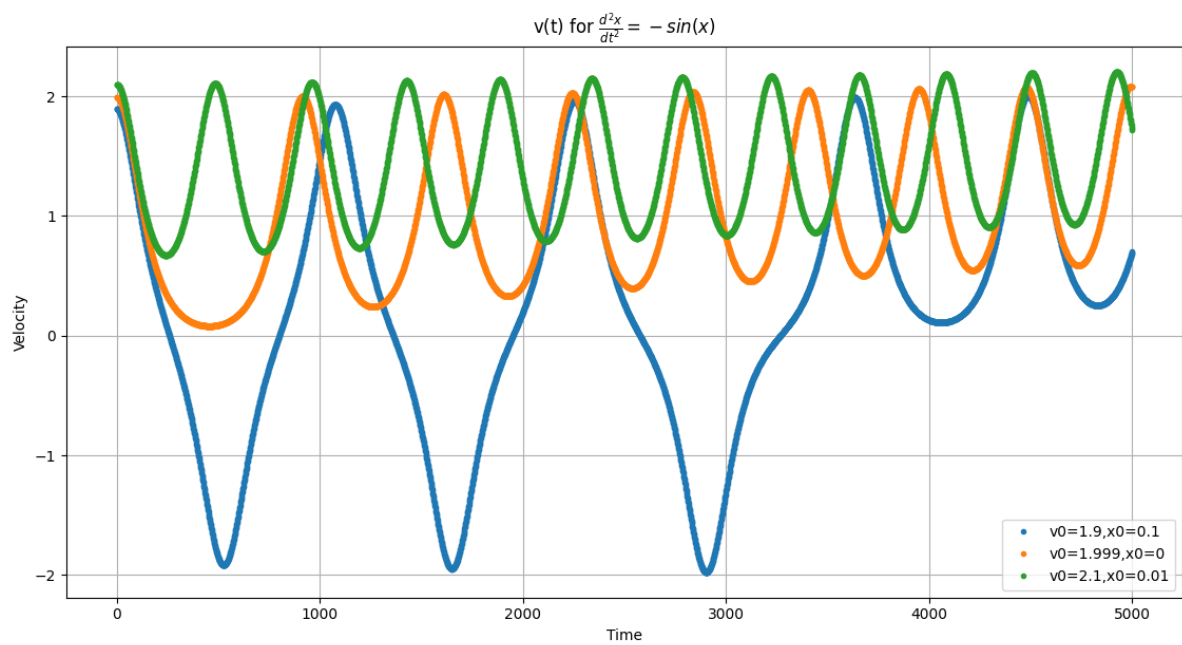
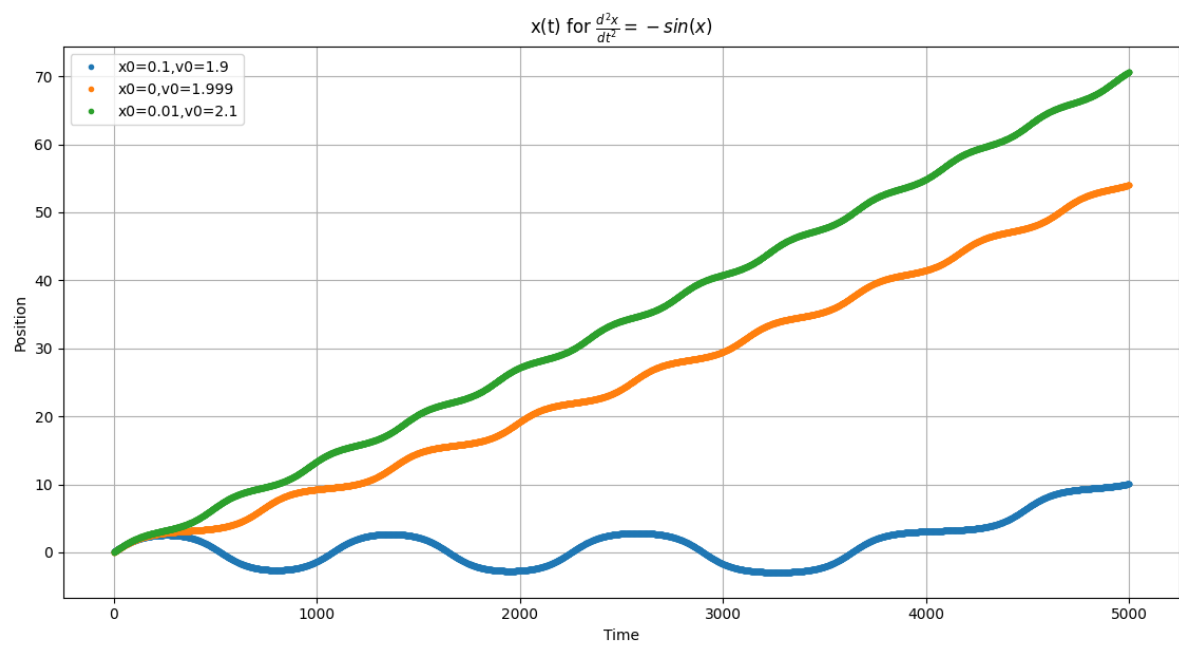


Q5:

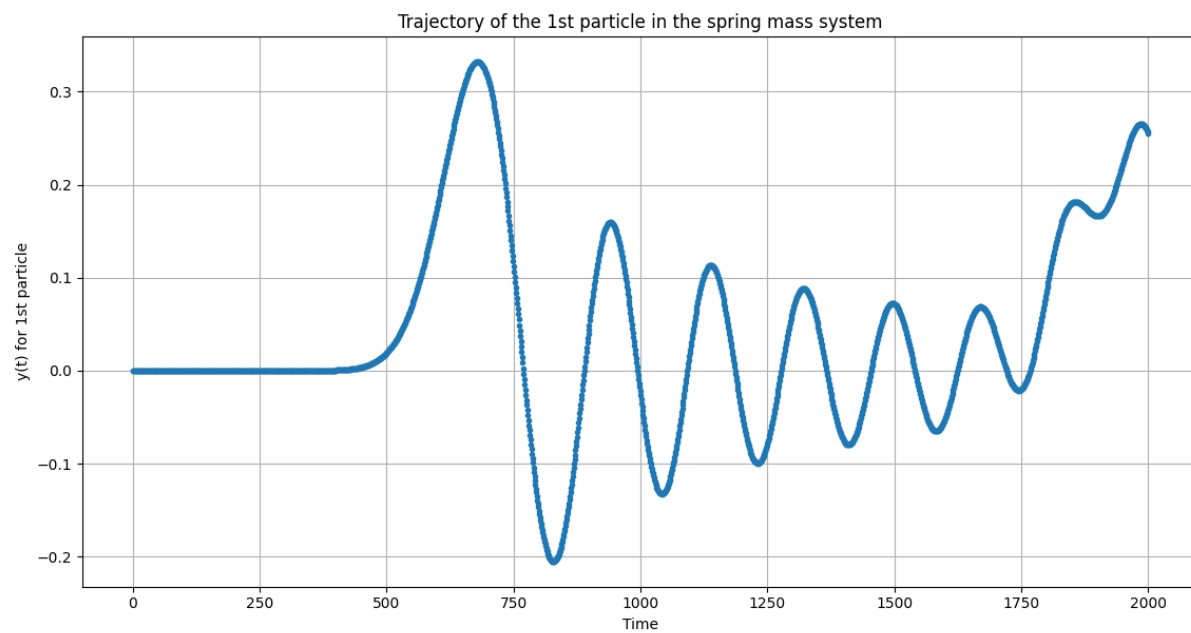
x after 50s(5000 iterations): 10.097020 (for $x_0=0.1$, $v_0=1.9$)

x after 50s(5000 iterations): 54.034934 (for $x_0=0$, $v_0=1.999$)

x after 50s(5000 iterations): 70.657876 (for $x_0=0.01$, $v_0=2.1$)



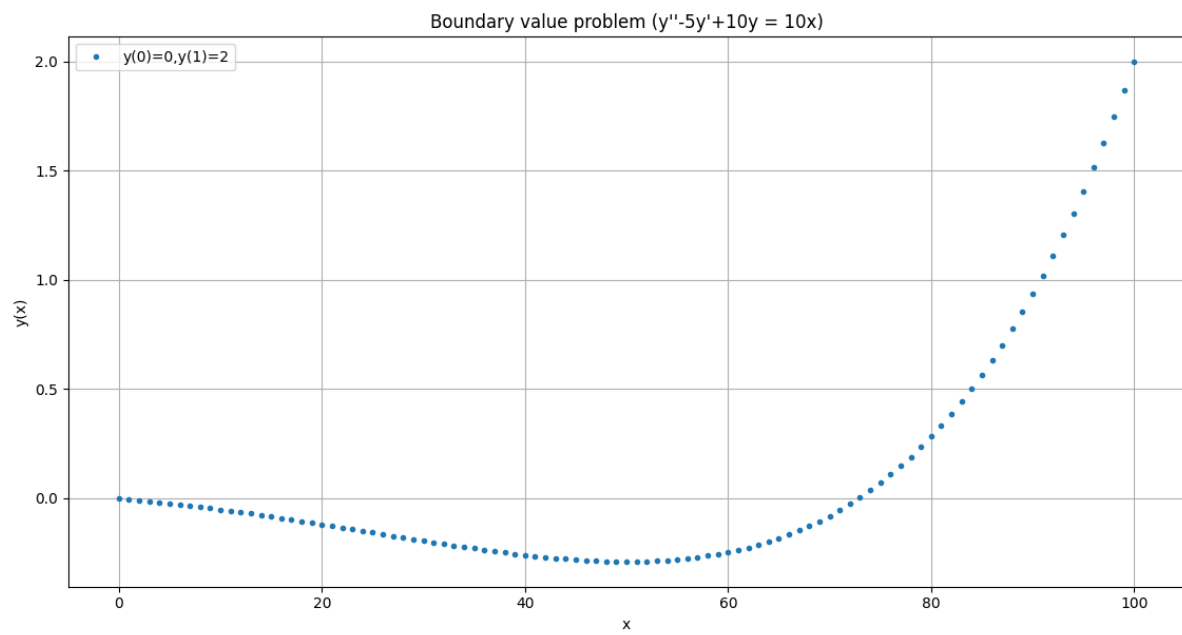
Q8:



Position of 1st particle after 2000 iterations: 0.2569

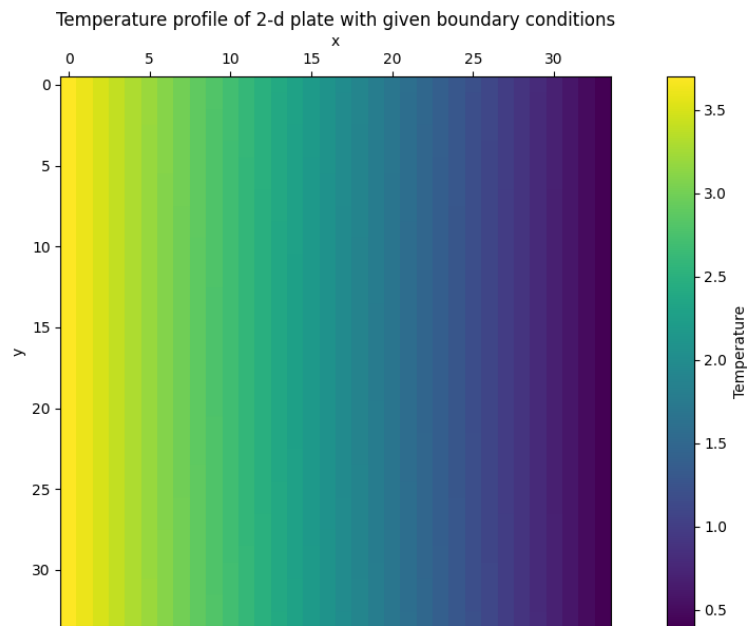
Initial conditions: $v_i = 0$, $y_1 = y_{26} = 0.8$

Q9:



Value of y at $x=0.8$: 0.233779

PDE Q1:



PDE Q2:

